

WASP-135b

R-1137

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_210505 · created 2026-07-30T23:39:17+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459339.8281 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1754 +/- 0.0077
Transit depth (Rp/Rs)^2	3.08 +/- 0.27 [%]
Orbital Inclination (inc)	82.69 +/- 1.07
Airmass coefficient 1 (a1)	0.957 +/- 0.016
Airmass coefficient 2 (a2)	0.04 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.64 %
Best Comparison Star	#2 - [405, 472]
Optimal Aperture	3.4
Optimal Annulus	9.28
Transit Duration (day)	0.076 +/- 0.0063

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1754 ± 0.0077 vs 0.139 (deviation 4.11σ)
Duration fit vs published	0.076 d vs 0.0687 d (deviation 1.0σ)
Mid-time O–C	-6.0 min
Depth SNR	19.8
Residual scatter	1.64 %

Light curve

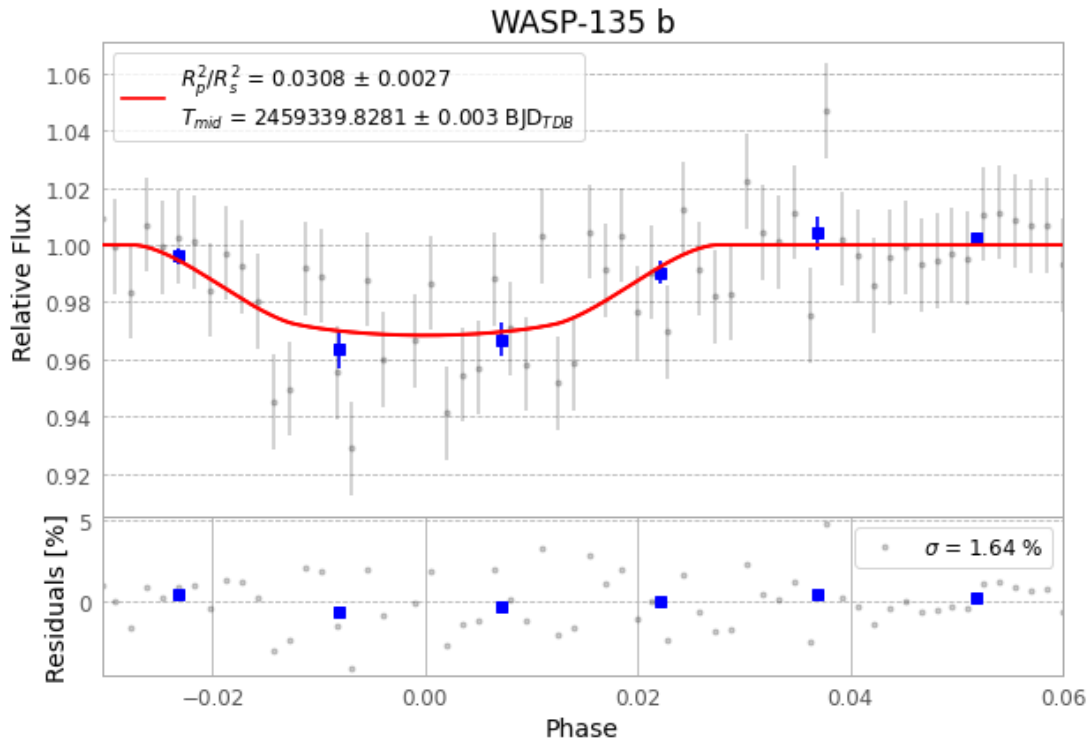


Figure 1 — FinalLightCurve_WASP-135 b_2021-05-04.png (SHA-256 044f0e159f8a25a4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [353, 223], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 95db42b1eca5ffd4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

62 FITS frames (41 MB), WASP-135210505064543.FITS → WASP-135210505094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-210505.log (138f46fbc48f...), FinalLightCurve_WASP-135 b_2021-05-04.png (044f0e159f8a...), FinalLightCurve_WASP-135 b_2021-05-04.csv (cfba97e836c0...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 23:39Z.